Math for CS & AI: Homework 7

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(Adapted from earlier versions of this course taught by Andrew Yao)

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Study Tips

In the lectures on complex analysis, I intentionally skipped some technical details and focused on the high-level ideas. To fill in the missing pieces, you should read the lecture notes carefully.

An effective way to study is to **alternate between reading and doing exercises**: read the notes for a while, then work on the corresponding exercises; then continue reading the notes for a while, then work on the corresponding exercises; repeat this process until you finish reading the lecture notes.

For this homework, you only need to read up to **Laurent series**.

PS: Starting from this homework, you will see more problems from **continuous mathematics** than from discrete mathematics. Some of you will be happy, and some will be sad. But you should learn to embrace continuous mathematics if you plan to work on AI. Be prepared for the change!

Problem Set

- Exercises 4.2–4.10 from Chapter 4 of the lecture notes:
  https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

Writing Guidelines

- Same as the first homework.
- We may deduct up to 3 points for poor writing.
- **Special Note:** Your submission should contain only a single PDF file. Please do not submit other files such as `*.zip` or `*.tex`.

Additional Hints

- We provide additional hints for some of the homework problems in the next few pages.
- If you get stuck on a problem, scroll down to find that problem and read the hints.
- Scroll *very slowly* so that you do not accidentally see hints for problems you have not thought about yet.
- Homework is meant to help you practice your problem-solving skills. It is your job to make sure you have thought enough before reading the hints. Ultimately, there will be no additional hints and no AI tools in the exams.
- Again, **peer learning is strongly encouraged**. Reading hints or getting help from AIs is not the same as learning from your classmate who has thought about the problem and can explain it to you. Make some friends who are willing to discuss problems or even have hotpot with you. Learn from their way of thinking, including both successful and unsuccessful attempts.

The rest of this document contains hints for the homework problems.



Think twice before scrolling down!

Hints

Problem 4.2

- There is a prefix sum in the recurrence.
- Read the discussion on prefix sums in the lecture notes.
- What is $\sum_{n\geq 1} nx^n$?
- Read the discussion on how to multiply the coefficient sequence elementwise by n.

Problem 4.3

- You can enumerate the subset that contains the element n.
- If this subset has k elements, then there are $\binom{n-1}{k-1}$ ways to choose the other $k-1$ elements.
- If you do not know how to proceed from here, read the example on summing variables with reordering.

Problem 4.4

- Read the proof of Euler's formula in the lecture notes.
- Prove the identities by comparing the coefficients of the power series on both sides.

Problem 4.5

- Read the discussion on why the complex conjugate is not differentiable at 0.
- So the trick is to approach a point z in different directions.
- Pick two directions that are easy to compute the limit in.

Problem 4.6

- For (a) and (b), you should just treat i as a normal variable, and do the calculation as if you were dealing with real calculus. Simplify the answer with $i^2 = -1$ in the end.
- For (c), if you do not know how to proceed, read the lecture notes more carefully!

Problem 4.7

- This problem is very tricky.
- Hint 1: $S_1 \cap S_2$ may not be connected.
- Hint 2: Go over the special functions that we have learned so far, including z^n, power series, rational functions, exponential function, principal logarithm, etc.

Problem 4.8

- If you do not know how to proceed, read the lecture notes more carefully!
- There is a direct theorem in the lecture notes that you can use. Find the nearest singularity of the function!

Problem 4.9

- Write down the power series expansion of e^{-t^2} at 0 based on the Taylor expansion of e^x.
- Integrate term by term to get the power series expansion of $\operatorname{erf}(x)$ on the real axis.
- Use Cauchy–Hadamard theorem to find the radius of convergence.

Problem 4.10

- If you do not know how to proceed, you are not understanding Cauchy's integral formula and Taylor series expansion deeply enough. Read the lecture notes more carefully!
- For (a), split the area between C_r and C_ϵ into two holomorphic parts.
- For (b) and (c), read the proof of Taylor series expansion.
- For (d), integrate term by term.
